

Ruonan Tian

Ph.D student in Bioinformatics
ZJU-UoE Institute, Zhejiang University, China
GitHub: <https://github.com/RuonanTian>
Institutional email: ruonan.23@intl.zju.edu.cn
Personal website: <https://ruonantian.github.io/rolan>

EDUCATION

Zhejiang University (ZJU-UoE Institute), Jiaxing, China

- Master student in Bioinformatics Sep 2021 – Jun 2023
 - **Supervisor:** Dr. Wanlu Liu
- Ph.D. student in Bioinformatics Sep 2023 – **Present**
 - **Supervisor:** Dr. Wanlu Liu
 - **Thesis:** Development and Application of High-Resolution Spatial Multi-Omics Methodologies

Huazhong Agricultural University, Wuhan, China

- Bachelor in Bioinformatics Sep 2017 – Jun 2021
 - College of Informatics
 - **GPA:** 3.76 / 4.00

RESEARCH INTERESTS

I am interested in multi-omics data analysis and computational framework development, with a focus on spatial transcriptomics and spatial metabolomics. My research aims to develop high-resolution spatial TCR technologies and integrated spatial multi-omics approaches to decode cellular and immune heterogeneity in intestinal tissues.

PUBLICATIONS

Selected Publications, * first or co-first authors # corresponding authors

- [1] **Tian, R.***, Xue, Z.*, Chen, Y., Qi, Y., Zhang, J., Yuan, J., Ruan, D., Lin, J., Liu, J., Wang, D., Ye, Y., Liu, W.#. Integrating Cross-Sample and Cross-Modal Data for Spatial Transcriptomics and Metabolomics with SpatialMETA. **Nature Communications** **16**, 8855. (2025)
- [2] **Tian, R.***, Yu, Z.*, Xue, Z., Wu, J., Wu, L., Cai, S., Gao, B., He, B., Zhao, Y., Yao, J., Lu, L., Liu, W.#. Evaluation of T Cell Receptor Construction Methods from scRNA-Seq Data. **Genomics, Proteomics & Bioinformatics** **22**, qzae086. (2024)
- [3] **Tian, R.***, Xue, Z.*, Ruan, D.*, Chen, P., Xu, Y., Dai, C., Shen, W., Ouyang, H.#, Liu, W.#, Lin, J.#. MSdb: an integrated expression atlas of human musculoskeletal system. **iScience** **26**, 106933. (2023)
- [4] Zhang, J.*, **Tian, R.***, Liu, J.*, Yuan, J.*, Zhang, S., Chi, Z., Yu, W., Yu, Q., Wang, Z., Chen, S., Li, M., Yang, D., Hu, T., Deng, Q., Lu, X., Yang, Y., Zhou, R., Zhang, X., Liu, W.#, Wang, D.#. A two-front nutrient supply environment fuels the physiology of the small intestine through differential regulation of nutrient absorption and host defense. **Cell**, **187**, 6251-6271.e20. (2024).
- [5] Ruan, D.*, **Tian, R.***, Li, Z., Yin, L., Tian, W., Chen, X., Xu, Y., Meng, Q., Zhou, W., Liu, H., Cai, H., Wang, X., Luo, Q., Lin, J.#, Liu, W.#, Shen, W.#. A Multi-Omics Atlas of Human MSCs Reveals Therapeutic Heterogeneity across Tissues and Clones. *Under review in Cell Stem Cell*.

Co-author Publications,

- [6] Chen, Y.*, Yan, Y.*, **Tian, R.**, Sheng, Z., Li, L., Chen, J., Liao, Y., Wen, Y., Lu, J., Liu, X., Sun, W., Wu, H., Liao, Y., Zhang, X., Chen, X., An, C., Zhao, K., Liu, W., Gao, J., Hay, D. C., Ouyang, H.#. Chemically programmed metabolism drives a superior cell fitness for cartilage regeneration. **Science Advances**, **10**, eadp4408. (2024)
- [7] Xue, Z.*, Wu, L.*, **Tian, R.**, Gao, B., Zhao, Y., He, B., Sun, D., Zhao, B., Li, Y., Zhu, K., Wang, L., Yao, J.#, Liu, W.#, Lu, L.#. Integrative mapping of human CD8+ T cells in inflammation and cancer. **Nature Methods**, **22**, 435–445. (2024)
- [8] Xue, Z.*, Wu, L.*, Gao, B.*, **Tian, R.**, Chen, Y., Qi, Y., Dong, T., Liu, Z., Bai, Y., Sun, D., Guo, Y., Chen, P., Zhao, Y., He, B., Wang, L., Yao, J., Lu, L.#, Liu, W.#. Disease associated human TCR characterization by deep-learning framework TCR.Deepinsight: A pan-disease single-cell TCRαβ repertoire reference atlas. **Cell Discovery**, **11**, 82. (2025)
- [9] Xiang, Y.*, Qian, Y.*, Li, Z.*, Wang, J.*, **Tian, R.**, Meng, W., Bu, J., Huang, F., Ai, Z., Wu, D., Chen, X., Wu, Y., Shen, L., Chan, Y.-S., Gao, Y., Ma, J., Liu, W.#, Gao, S.#, Zhang, D.#, Liang, H.#. Endogenous retroviruses synthesize heterologous chimeric RNAs to reinforce human early embryonic development. **Science** **391**, eadv5257. (2026)

In preparation,

Tian, R.*, Chen, Y., Qi, Y., Xue, Z., Liu, W.# ScSpaTCR: An Integrated Platform for Single-Cell Resolution Spatial Sequencing of T Cell Receptor Repertoires (In preparation)

COPYRIGHTS

- **Copyrights of computer software**
 - Lin, J., **Tian, R.**, Xue, Z., Ruan, D., Liu, W. An analysis and visualization software for musculoskeletal multi-modality data (2023SR1142549)
 - **Tian, R.**, Xue, Z., Liu, W. A deep-learning software for integrated analysis of spatial transcriptomics and spatial metabolomics [SpatialMETA] v1.0. Registration No.: 2025R11L2933267, under review.
- **Patents**
 - Liu, W., **Tian, R.**, Xue, Z. A deep-learning-based method for integrating spatial transcriptomics and spatial metabolomics. Patent Application No.: 2025110969999. Status: Published and entered substantive examination.

AWARDS & SCHOLARSHIPS

- **National Scholarship** 2025
- **Zhejiang University PhD Student "Qiusi New Star" Program** 2024
- **National Scholarship** 2023
- **Outstanding Graduate Student, Zhejiang University** 2023
- **Outstanding Graduate, Huazhong Agricultural University** 2021
- **Gold iGEM (as team advisor, leader, member)** 2020, 2019, 2018
- **First Prize, The Bioinformatics Design and Skills Competition** 2020, 2019
- **Top 10 Stars of Entrepreneurship and Innovation in the University** 2020

CONFERENCES & PRESENTATIONS

- **GPB2025 (Haining, China)** 2025
 - **Oral presentation:** Integrating cross-sample and cross-modal data for spatial transcriptomics and metabolomics with SpatialMETA
- **IMMUNOLOGY2025 (Honolulu, USA)** 2025
 - **Poster:** High-Resolution Spatial Multi-Omics for Unraveling Immune Heterogeneity in Inflammatory Bowel Disease
- **Human Cell Atlas Asia Meeting 2024 (Hong Kong, China)** 2024
 - **Poster: SpatialMETA:** A Novel Framework for Integrating Spatial Transcriptomics and Metabolomics Data

REVIEWER

- **Ad hoc reviewer:** *Nature Methods (co-reviewer), Genomics, Proteomics & Bioinformatics*

SKILLS

- **Programming**
 - Python (NumPy, Pandas, Django, PyTorch, Scanpy, etc.), R (Seurat, ggplot2, etc.), HTML/CSS, JavaScript, SQL, etc.
- **Dry Experiment Skills**
 - Multi-omics data analysis (Spatial multi-omics, scRNA-seq, scTCR-seq, Hi-C, ChIP-seq, ATAC-seq, RNA-seq etc.), Algorithm development, Web development
- **Wet Experiment Skills**
 - T/B cell isolation, Molecular cloning, ATAC-seq library, Spatial TCR library, etc.
- **Miscellaneous**
 - Linux, Adobe Illustrator, Microsoft Office, LaTeX, Git, etc.